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An exact likelihood analysis of the multinomial probit model

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Abstract

We develop new methods for conducting a finite sample, likelihood-based analysis of the multinomial probit model. Using a variant of the Gibbs sampler, an algorithm is developed to draw from the exact posterior of the multinomial probit model with correlated errors. This approach avoids direct evaluation of the likelihood and, thus, avoids the problems associated with calculating choice probabilities which affect both the standard likelihood and method of simulated moments approaches. Both simulated and actual consumer panel data are used to fit six-dimensional choice models. We also develop methods for analyzing random coefficient and multiperiod probit models.

Key words: Multinomial probit model; Gibbs sampling; Bayesian analysis

JEL classification: C11; C25; C35

1. Introduction

The multinomial probit model (MNP) was first formulated in the econometrics literature by Hausman and Wise (1978) and Daganzo (1980) (see, also, the early work of Thurstone, 1927). The appeal of the multinomial probit model is the relaxation of the independence of irrelevant alternatives property of multinomial logit models. In addition, the multinomial probit model is closely

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related to a multiperiod probit model in which a binary choice decision is observed over multiple time periods with correlated errors. Until recently, however, applied researchers have rarely used the MNP due to computational difficulties in computing the maximum likelihood estimates (MLE). An alternative suggested by McFadden (1989) is to use a method of simulated moments (MSM) approach which avoids direct evaluation of the likelihood function (see, also, Keane, 1993); the method of simulated scores (Hajivassiliou and McFadden, 1990) also avoids direct evaluation of the likelihood function. However, both the MLE and MSM approaches require a method for evaluation of choice probabilities which, for the MNP model, involves $(p - 1)$ -dimensional integrals of a multivariate normal distribution over a cone where p is the number of choice alternatives. In the case of the MLE, extremely accurate estimates of these choice probabilities are required (Borsch-Supan and Hajivassiliou, 1990). Even the MSM approach is sensitive to the method used to estimate the probabilities.

The MLE and MSM approaches rely on asymptotic approximations to the sampling distribution of the estimators. The difficulty in fitting the MNP model renders a parametric bootstrap approach to computing some sort of approximate finite sample distribution computationally impractical. Due to the discrete nature of the dependent variable, a much larger sample size may be required for accurate asymptotic approximations than needed to apply asymptotics to standard continuous dependent variable models.

Previous work on Bayesian analysis of choice models includes Zellner and Rossi (1984), Koop and Poirier (1990), Carlin and Polson (1991), and, of particular relevance to this paper, Albert and Chib (1993). In this paper, we present the first feasible approach to Bayesian analysis of the MNP model. We construct a Markov chain-based sampler which can be used to draw directly from the exact posterior distribution and perform *finite* sample likelihood inference to any desired degree of accuracy. By drawing sequentially from a series of conditional distributions, a sequence of draws can be made which converge in distribution to a draw from the posterior. Our approach avoids direct evaluation of the likelihood and the associated problems of approximating choice probabilities. Results on the properties of general state space Markov chains can be specialized to prove strong uniform and geometric ergodicity properties which justify the use of the sampler to compute any desired quantity of interest such as posterior moments and probability intervals.

We demonstrate the usefulness of our procedure using both simulated and actual consumer data. We implement a Bayesian random coefficient or hierarchical model using our MNP sampler and apply this model to consumer panel data. We also develop procedures to analyze the multiperiod probit model as well as variance component models.

2. The MNP model and Bayesian analysis

We observe N independent observations on the multinomial vector, y_i , with p choice alternatives. Consider a standard latent variable interpretation of the multinomial probit model. For each choice occasion, a latent normal vector $z_i(p \times 1)$ is present. Choice alternative j is observed if the j th component of z_i is larger than all other components.

$$z_i = R_i \beta + u_i, \quad u_i \sim N(0, V), \quad (1)$$

$$y_{ij} = 1 \quad \text{if} \quad z_{ij} \geq \max(z_i),$$

$$= 0 \quad \text{otherwise.}$$

R_i is a $p \times k$ matrix of regressor values for the i th observation for each alternative. The z_i can be stacked up into a multiple regression with correlated errors, $z = R\beta + u$, where $z' = [z_1', \dots, z_N']$, $R' = [R_1', \dots, R_N']$, and $u' = [u_1', \dots, u_N']$. Throughout this paper, we take a strictly parametric approach by assuming a specific error distribution and form for the latent variable regression function.

Each of the choice probabilities is the integral of a multivariate normal distribution over a cone. The probability of choice j on observation i , denoted Pr_{ij} , is the $Pr(z_{ij} \geq \max(z_{ik}), k = 1, \dots, p)$. We follow the usual practice of expressing each z_{ij} relative to z_{ip} by letting $w_i' = (w_{i1}, \dots, w_{i,p-1})$ with $w_{ij} = z_{ij} - z_{ip}$. Using the w_i parameterization, we can write the latent variable system as a $(p-1)$ -dimensional multivariate regression, $w_i = X_i \beta + \varepsilon_i$, $\varepsilon_i \sim N(0, \Sigma)$. X_i denotes the $(p-1) \times k$ matrix of observation vectors obtained from the original R_i by subtracting the p th row from the first $(p-1)$ rows. For notational simplicity, we can collapse the y_i multinomial vector into a simple indicator variable, $d_i = 0, \dots, p-1$, which indicates the choice. The choice model is that $d_i = 0$ if $w_i < 0$; $d_i = \text{index of } \max(w_{ij}, j = 1, \dots, p-1)$ otherwise. These choice conditions reveal the fundamental identification problem in the MNP. Scaling the w_{ij} by a positive constant, c , doesn't change the value of d_i .

A Bayesian analysis of the MNP model requires the specification of a prior over the parameters (β, Σ) and computation of the posterior density.

$$p(\beta, \Sigma | y_1, \dots, y_N, X) \propto p(\beta, \Sigma) l(\beta, \Sigma). \quad (2)$$

The MNP likelihood is formed by the product of the N independent multinomial distributions, $l(\beta, \Sigma | y_1, \dots, y_N, X) = \prod_{i=1}^N \prod_{j=1}^p Pr_{ij}^{y_{ij}}$. As for the prior, it is convenient to employ a normal prior on β , $\beta \sim N(\bar{\beta}, A^{-1})$ and an independent Wishart prior on Σ^{-1} (denoted G below),

$$p(\beta | \bar{\beta}, A) \propto |A|^{0.5} \exp \left\{ -\frac{1}{2} (\beta - \bar{\beta})' A (\beta - \bar{\beta}) \right\} \quad (3a)$$

and

$$p(G | v, V) \propto |G|^{(v-p-1)/2} \text{etr}\{-\frac{1}{2}GV\}. \quad (3b)$$

The diffusion of the prior relative to the likelihood determines how strong an influence the prior will have on the posterior distribution of the identified model parameters. The situation in which the investigator has no strong prior beliefs about the location of the model parameters may be approximated by choosing extremely diffuse, but proper, priors. It is a simple matter to specify very diffuse priors by taking the precision matrix of β to be very small and by setting v to a very small number. For very diffuse priors on β , the choice of the location of the prior, $\bar{\beta}$, and the correlation structure in A is not critical. Similarly, the Wishart prior on G can be regarded as arising from an imaginary previous sample with sample size v . By setting v to a small number relative to the actual sample size used in the analysis, we can insure that the prior on G remains diffuse relative to the likelihood.

These proper priors will insure that the posterior is proper even though the likelihood is flat on a lower-dimensional manifold. The main consequence of the identification problem is that a certain marginal from the posterior will reflect solely information imparted by the prior. Other posterior marginals, such as $\beta/(\sigma_{11})^{0.5}$ and σ_{ij}/σ_{11} , will be determined by both prior and sample information. Our approach will be to use informative but very diffuse natural conjugate priors and report posterior marginals of properly normalized parameters. In this way, our posterior results will primarily reflect sample information. In Section 4, we discuss other important issues which arise in connection with the use of diffuse priors and Gibbs samplers for unidentified models.

3. A Gibbs sampling algorithm for the MNP

Because direct evaluation of the MNP likelihood is very costly, standard numerical methods for conducting Bayesian analysis such as importance sampling are infeasible. An alternative to integration of the posterior is a direct simulation approach in which draws are made from the posterior and used to form empirical estimates of distributions or sample mean estimates of integrals. While direct simulation from the posterior is not possible, an indirect method called the Gibbs sampler can be used which relies only on drawing from various conditional distributions. The Gibbs sampler introduced by Geman and Geman (1984) and Tanner and Wong (1987) relies on the remarkable result that iterative, recursive sampling from conditional distributions results in a sequence of random variables which converge in distribution to the joint distribution. Gelfand and Smith (1990) provide a general discussion of the use of the Gibbs sampler (see, also, Casella and George, 1992). Hajivassiliou and McFadden (1990) consider the use of Gibbs sampling methods to simulate logarithmic

derivatives of rectangular probability integrals. Our approach avoids direct simulation or approximation of choice probabilities and exploits the full latent variable structure of the MNP model. In Section 5, we analyze both the theoretical and practical convergence properties of the MNP sampler.

The basic idea behind the Gibbs sampler is that it is possible to draw from the joint distribution of a collection of random variables by drawing successively from various conditional distributions. To make this point clear, consider the bivariate case. We wish to draw from the joint distribution of $\theta = (\theta_1, \theta_2)$ using only our ability to draw from the conditional distributions of $\theta_1|\theta_2$ and $\theta_2|\theta_1$. We start from any point in the support of the joint distribution and draw iteratively from the conditional distributions, each in turn. That is, to get from θ to θ' we draw $\theta'_1|\theta_2$ and $\theta'_2|\theta'_1$. The sequence $\{\theta^k\}$ converges in distribution to a draw from the joint distribution. If we plot the sequence of actual draws in the parameter space, these draws might start far from the center of mass for the joint distribution, but they will converge to a stationary Markov chain which navigates through the regions of the parameter space which are accorded high probability by the joint distribution. More generally, we break our parameters into k groups and cycle through these k conditional distributions. The key to successful application of the Gibbs sampler in this and many other problems is the availability of conditional distributions which can be sampled from at low cost.

In the analysis of probit models, the introduction of the latent variables, w , in what is known as the data augmentation step, makes the application of a Gibbs sampler feasible. In analysis of discrete dependent variable models, Albert and Chib (1993) were the first to note the usefulness of data augmentation.¹ *Conditional on w* , the MNP model is simply a standard Bayesian linear model set-up. After adding the w vector to the set of variables in the data augmentation step, we can define a Gibbs MNP sampler via a sequence of conditional distributions which define a Markov chain that progresses from (w, β, G) to (w', β', G') in each step. Our strategy will be to construct draws from the $w, \beta, G|d$ distribution via the Gibbs sampler and concentrate on the marginal of (β, G) , $p(\beta, G|d) = \int p(w, \beta, G|d)dw$, where $G = \Sigma^{-1}$, $d' = (d_1, \dots, d_N)$, and $w' = (w'_1, \dots, w'_N)$.

It is a simple matter to draw from $\beta|w, G$, and $G|w, \beta$ using the well-known normal posterior distribution for $\beta|w, G$ and the Wishart posterior of $G|w, \beta$ (see Zellner, 1971, for details). Note that the conditioning argument, d , is redundant given w . The only challenging part of this Gibbs strategy is to devise a method of drawing from the conditional of $w|\beta, G$. We recognize that because the observations are independent we can draw each of the w_i separately and piece them together. The distribution of $w_i|\beta, G, d_i$ is a $(p - 1)$ -variate normal distribution

¹ See, also, Carlin and Polson (1991).

truncated over the appropriate cone in R^{p-1} . If $y_{ij} = 1$, then $w_{ij} > \max(w_{i,-j}, 0)$; if $y_{ij} = 0$, then $w_{ij} < \max(w_{i,-j}, 0)$. $w_{i,-j}$ is a $(p-2)$ -dimensional vector of all of the components of w_i , excluding w_{ij} . We avoid this problem of directly drawing from the truncated multivariate normal by using the conditional distributions of $w_{ij}|w_{i,-j}, \beta, G, d_i$. If we draw, in turn, from all of the conditional distributions of each component of the w_i vector given all of the others, we have constructed a Gibbs sampler which can generate sequences which converge to a draw from the truncated multivariate normal distribution. Albert and Chib (1993) suggest that truncated multivariate normal draws be made by a direct accept/reject method in which draws are made from the normal distribution and discarded if they do not fall in the appropriate cone.

Our MNP sampler constructs sequences of draws by sampling consecutively from the three groups of conditional posterior distributions. Recall that all posterior distributions are conditional on both d and X .

$$\text{a) } N(p-1) \text{ conditionals of } w_{ij}|w_{i,-j}, \beta, G, d_i, \quad i = 1, \dots, N, \quad (4)$$

$$j = 1, \dots, p-1,$$

$$\text{b) } \beta|w, G,$$

$$\text{c) } G|w, \beta.$$

$w_{i,-j}$ means the vector $(w_{i1}, \dots, w_{i,j-1}, w_{i,j+1}, \dots, w_{i,p-1})$. We start from the prior modes of β and G and set the first iterate of the w vector to 0.

The first step of the MNP sampler requires a draw of $w_{ij}|w_{i,-j}, \beta, G, d_i$. $w_{ij}|w_{i,-j}, \beta, G, d_i$ has a truncated *univariate* normal form with a truncation point easily calculated from the implied R^{p-1} regions and with moments computed from the conditional distribution of $w_{ij}|w_{i,-j}, w_{ij}|w_{i,-j}, \beta, G \sim N(m_{ij}, \tau_{ij}^2)$. The conditional mean and variance can be computed by conveniently partitioning the G matrix. For convenience, permute the rows and columns of G so that the j th row and column are the last,

$$G = \begin{bmatrix} \Sigma_{(-j)(-j)} & \sigma_{(-j)j} \\ \sigma_{j(-j)} & \sigma_{jj} \end{bmatrix}^{-1} = \begin{bmatrix} \Sigma_{(-j)(-j)}^{-1} + FE^{-1}F' & -FE^{-1} \\ -E^{-1}F' & E^{-1} \end{bmatrix}, \quad (5)$$

where

$$E = \sigma_{jj} - \sigma_{j(-j)} \Sigma_{(-j)(-j)}^{-1} \sigma_{(-j)j} = \tau_{ij}^2, \quad F = \Sigma_{(-j)(-j)}^{-1} \sigma_{(-j)j}.$$

$m_{ij} = x'_{ij}\beta + F'(w_{i(-j)} - X_{i(-j)}\beta)$. x_{ij} is the j th row of X_i and $X_{i(-j)}$ is the submatrix created by deleting the j th row from the X_i .

The truncation of the conditional distribution of w_{ij} comes from the fact that if $d_i = j$ then $w_{ij} > \max(w_{i,-j}, 0)$ whereas if $d_i \neq j$ then $w_{ij} < \max(w_{i,-j}, 0)$. The draws can be made given an algorithm for drawing from a standard normal distribution truncated from below. Our rejection strategy is to splice together three different methods for different truncation point regions. We use three

methods: 1) exponential rejection sampling (see Devroye, 1986), 2) half normal sampling, and 3) direct normal rejection sampling (see, also, Geweke, 1991).

The next step of the sampler requires a draw from conditional distribution b), the conditional distribution of β given w , G , and X . Conditional on G , it is a simple matter to transform the regression equation to the case of iid errors and apply standard conjugate prior theory. Let $G = CC'$. Our method of drawing G from the Wishart distribution provides the Cholesky root C at no additional expense. Transforming the equation by premultiplication by C' reduces the system to a system with $N(0, I)$ errors,

$$C'w_i = C'X_i\beta + C'\varepsilon_i, \quad (6)$$

$$w_i^* = X_i^*\beta + \varepsilon_i^*, \quad \varepsilon_i^* \sim N(0, I_{p-1}),$$

or in stacked form,

$$w^* = X^*\beta + \varepsilon^*, \quad \varepsilon^* \sim N(0, I_{N(p-1)}).$$

Thus, the conditional posterior distribution of β is normal,

$$\beta|w, G \sim N(\hat{\beta}, \Sigma_\beta), \quad \Sigma_\beta = (X^{*'}X^* + A)^{-1}, \quad \hat{\beta} = \Sigma_\beta(X^{*'}w^* + A\bar{\beta}). \quad (7)$$

d is a function of w so that it is not necessary to add d to the conditioning arguments.

The third conditional distribution of G given β , w , X can be obtained from standard Bayesian analysis of a covariance matrix and Wishart theory. Given β , w , and X , we actually observe $\varepsilon_i = w_i - X_i\beta$. We combine the conjugate Wishart prior with the likelihood to obtain a Wishart posterior,

$$G|\beta, w \sim W\left(v + N, V + \sum_{i=1}^N \varepsilon_i \varepsilon_i'\right). \quad (8)$$

Draws from a $W(v, S)$ distribution can be obtained by drawing from the standard Wishart distribution. Let $S^{-1} = LL'$.

$G = L\tilde{V}L'$, where $\tilde{V} \sim W(v, I)$.

$\tilde{V} = TT'$, where T is a lower triangular draw made by independently drawing the square root of χ^2 variates for the diagonal elements, $T_{ii} \sim \sqrt{\chi_{v-1+1}^2}$ and $N(0, 1)$ draws for off diagonal elements. Each draw of G is made by constructing a T draw and premultiplying by the lower triangular matrix, L . $G = CC'$, $C = LT$ (see Johnson, 1987, pp. 203–204).

Using the strategy outlined above, we draw in turn from each of the $N(p-1) + 2$ conditional distributions given by a), b), and c). These draws will converge to a single draw from the posterior distribution. This process is repeated to obtain additional draws from the posterior.

4. Identification and the behavior of the MNP sampler

As is well known, an identification problem arises in the MNP model because any location and scale shift will not change the observed choices (see Dansie, 1985; Bunch, 1991; Keane, 1990).² The standard solution to the random shift in location problem is to collapse the system by differencing with respect to one of the choice alternatives giving in our notation the system, $w = X\beta + \varepsilon$. Even in this system, the parameters $(c\beta, c^2\Sigma)$ for any $c > 0$ are observationally equivalent to (β, Σ) . In our Bayesian approach, we compute the full posterior over β and Σ and simply report the marginal posterior distribution of the identified parameters, e.g., β/σ_1 and error correlation coefficients. In the context of Markov chain Monte Carlo approaches to Bayesian analysis of unidentified models, improper priors should be avoided. With diffuse priors, the full posterior over the entire set of parameters will be improper. The consequence of this is that the Gibbs sequence of draws will follow a random walk which will not converge. While it may still be possible to prove that the sequence of properly normalized parameters generated by the Gibbs sampler follows a Harris recurrent ergodic chain, this has not yet been proven and will require a modification of the methods used here.

To illustrate this phenomenon in practice, consider the case of a binomial probit ($p = 2$) with only one variable and a slope coefficient ($k = 1$). A small dataset of 1000 observations was generated and a sequence of 2000 Gibbs draws was produced. The top half of Fig. 1 shows a scatterplot of the (β, σ) draws. The draws march up and down the line $\beta/\sigma = \text{true value}$. While β and σ vary over a wide range of values, the draws never stray too far from the identified ratio β/σ . If we project the draws onto the line and plot the distance of the projected points from the point $(2, \sqrt{2})$, we can see the random walk nature of the draws (see bottom half of Fig. 1).

Our approach to the identification problem is to use proper, but highly diffuse, priors, and report the posterior distribution of the properly normalized parameters. This keeps the Gibbs sequences from wandering too far a field in the full parameter space and allows application of the theory of Harris recurrent Markov chains. An alternative to putting a proper prior over the full set of parameters would be to condition on $\sigma_{11} = 1$ and construct a Gibbs sampler from the conditional posterior of $\Sigma|\sigma_{11} = 1$. This would take us away from the analytically convenient Wishart form which is useful in constructing the sampler.

A useful way of viewing our procedure based on a proper prior is that we are using the full parameterization as a device to assess a prior on the identified

²To see this, consider the full system $z = R\beta + u$. The transformation from z to v defined by $mi + cz, v' = (1, \dots, 1)$ will leave the choices unchanged. It is also true that the location shift can be random from observation to observation.

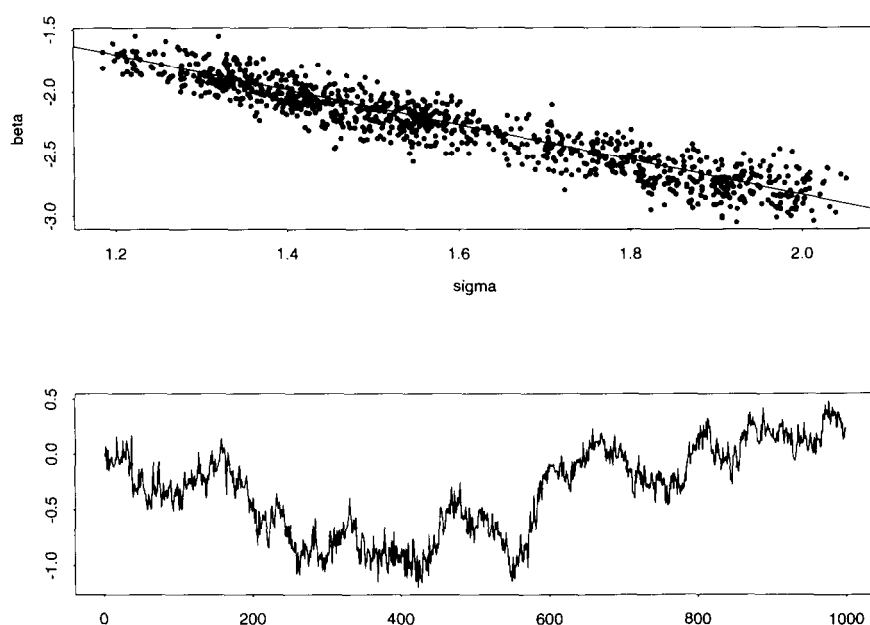


Fig. 1. The identification problem in the binomial probit.

parameters $\tau' = (\beta'/\sigma_1, \rho_{ij}, \sigma_j/\sigma_1, j = 2, \dots, p)$. It is, therefore, important to examine the induced prior distribution over τ to insure that it is appropriately diffuse or that it suits the investigators specific prior beliefs. Our Markov chain methods are used to compute the marginal posterior over τ , $\pi(\tau|\text{data}) = \int \pi(\sigma_1, \tau|\text{data})d\sigma_1$. Marginalizing over the posterior will yield identical results as an analysis in which we start by marginalizing over the full prior to obtain the prior on τ and then proceed to compute the posterior of τ . This is true because the likelihood depends only on the identified parameters, τ . We choose to marginalize over the posterior because this avoids computation of the likelihood over the identified parameters. Thus, the fact that σ_1 and τ are a priori dependent poses no logical difficulties since we use only the marginal prior on τ in our analysis.

An alternative method of identification would be to fix one of the elements of β at some value ($\beta_s = 1$). This approach has the advantage that it is easily implemented in our algorithm and that improper, diffuse priors could be used. When drawing β_{-s} and Σ , we would use $w - X_s\beta_s = X_{-s}\beta_{-s} + \varepsilon$, where X_{-s} is the X matrix without the s th column and β_{-s} is the β vector without the s th component. The draw of w would be the same. The approach has the disadvantage that it requires that there is one coefficient whose sign may be determined

a priori (e.g., a price coefficient). Another possible disadvantage of this approach is that it may be difficult to assess priors on the free parameters conditional on $\beta_s = 1$.

This alternative approach amounts to putting a degenerate informative prior on β_s which achieves model identification. An extension of this idea suggested by a referee would be to put a nondegenerate informative prior on β_s . In particular, it would be simplest (although not necessary) to make β_s have a Bernoulli distribution with support points -1 and 1 . The Gibbs sampler could then be enlarged to compute the joint posterior over all of the parameters including β_s . To conduct a meaningful posterior analysis of the other regression coefficients it would then be necessary to normalize with respect to β_s . Without the normalization, we would obtain a noninterpretable marginal posterior which would be a mixture of the conditional posteriors given $\beta_s = 1$ and $\beta_s = -1$. This mixture is not interpretable because the model parameters only have meaning relative to a sign normalization and we are mixing over two different sign normalizations. Thus, the procedure in which a proper Bernoulli prior on β_s is combined with a normalization should be viewed as a special case of our general procedure in which at least some of parameters have a proper prior and the posterior is renormalized.

5. Convergence properties of the MNP algorithm

5.1. Theoretical properties

A burgeoning statistical literature uses results on general state space Markov chains (the key reference is Nummelin, 1984) to prove various convergence results regarding the Gibbs sampler and related Markov chain Monte Carlo methods. In this section, we discuss the properties of the MNP sampler using results from this literature and relate a new result on geometric convergence for the binomial case. For an excellent discussion of the theoretical properties of Gibbs samplers and other related Markov chain methods see Tierney (1991).

In practice, the MNP sampler will be started from arbitrary points in the state space and iterated until it converges to the stationary or invariant distribution of the chain. The marginal distribution of (β, G) of the invariant distribution of the chain should be the posterior distribution of the model parameters. Therefore, it is important to establish that the chain converges to a unique invariant distribution which is the posterior from any point in the state space. Any chain defined by a strictly positive transition kernel such as ours (product of truncated and untruncated normal and Wishart distributions) is irreducible and aperiodic and, hence, has a unique invariant distribution which is the posterior (see, for example, Tierney, 1991, Theorem 1 and Corollary 2, Theorem 2, p. 10). It is also possible to establish ergodicity of the chain so that a law of large numbers and

central limit theorem can be used to make inferences about posterior quantities using the Gibbs draws.

One common practice which we follow in our empirical applications is to discard the first T^* draws under the assumption that the chain will have converged to within a very close tolerance to the invariant distribution. The size of T^* is determined by the rate of convergence of the MNP sampler. Under certain conditions, it is possible to show that the Gibbs sampler converges at a geometric rate to the invariant distribution. Different sufficient conditions for this rapid rate of convergence are given in, for example, Roberts and Polson (1990), Tierney (1991), and Schervish and Carlin (1992). These sufficient conditions, such as the minorization condition introduced into the Markov chain literature by Nummelin, can be extremely difficult to verify for a given chain. In the Appendix, we prove geometric convergence of the MNP sampler for the binomial case. Albert and Chib (1993) note the fast convergence of their Gibbs sampler for the binomial probit case but do not demonstrate that their binomial Gibbs sampler is convergent or geometrically ergodic.

5.2. *Practical convergence issues*

The theoretical literature offers only qualitative guidance into the determinants of the rate of convergence. Experience with the Gibbs sampler has shown that, for the most part, the sampler converges very rapidly, far more rapidly than the theoretical bounds would indicate. This suggests that the bounds are not very tight, especially in the high-dimensional state spaces associated with models involving latent variables. It is, therefore, important to conduct an empirical analysis of convergence in our application to the MNP problem. There are two important considerations: 1) how long must the Gibbs sequence be run in order to be confident that the effect of initial conditions have dissipated? and 2) what is the information content of a given sequence of Gibbs draws from the stationary distribution?

To answer the first question, we conduct an analysis of the sensitivity of estimated posterior distributions to various, widely dispersed initial conditions (see Gelman and Rubin, 1992a, for a discussion of convergence issues). For each of a large set of very different initial conditions, we run out the Gibbs sampler for fairly large number of draws (in our examples, we used 5000 draws). We then compare the simulated marginal posterior distributions for each of the last few thousand draws for each of the 'short' Gibbs runs. If these estimated posterior distributions show little variation relative to the starting points, we can feel reasonably confident that the Gibbs sampler will converge to the stationary distribution within the specified number of draws. In our experience, the MNP Gibbs sampler converges very rapidly (within only a few hundred draws) to the stationary distribution. Another possible benefit of this sensitivity analysis can be the detection of a multimodal likelihood (this cannot be ruled out in the

MNP model, although the binomial probit log likelihood is globally concave). If the likelihood were to have modes of similar heights, we might see the Gibbs draws 'attracted' to different modes depending on the starting values. In a long enough run, all modes should be visited with probability 1 even though the Gibbs sequence might jump from mode to mode. Gelman and Rubin (1992b) suggest convergence diagnostics based on sampling initial conditions from an overdispersed multimodal approximation to the posterior. In our experience with the MNP model, there has been no evidence of a multimodal likelihood. In our problem, the initial conditions involve not only the parameters but the latent variables, w , rendering the Gelman and Rubin strategy impractical.

Having satisfied ourselves that the MNP sampler converges to the stationary distribution in T^* draws, we then make one long run of length T^{**} and keep the last $T = T^{**} - T^*$ draws. The Gibbs sequence of draws forms a stationary and ergodic time series and we can use the theory of stationary time series processes to deduce the accuracy Gibbs based estimates of posterior quantities. The MNP Gibbs sequences generate a highly nonlinear time series process with a complicated autocorrelation pattern and conditional heteroskedasticity.

For posterior quantities which can be expressed as posterior moments, our estimates based on Gibbs draws provide an ideal application for standard methods of inference developed in the time series literature. Our advantage over traditional users of these techniques is that we can prove stationarity and ergodicity which must be assumed for economic time series and we can control the sample size at will. The accuracy of our Gibbs estimates is only limited by the analyst's patience and computing resources. However, for a given allocation of computing time, it is important to determine the accuracy of the Gibbs estimates. To illustrate how the standard results can be applied to measure the accuracy of MNP Gibbs estimates, consider the estimation of $\tau = E[g(\theta)]$. Given a sequence of T draws, it is natural to use the method of moments estimator, $\hat{\tau} = \Sigma g(\theta_i)/T$. Standard spectral density methods using Bartlett weights can be used to provide a consistent and positive definite estimate of the asymptotic variance-covariance matrix of $\hat{\tau}$ (see, for example, Newey and West, 1987; Gallant, 1987; Geweke, 1992).

6. Some simulated examples

To illustrate the structure of the MNP and the functioning of our algorithm, we consider some simulated examples. We also want to check the speed of convergence, sensitivity of the sampler to initial conditions, normality of the posterior distributions, and investigate the correlation structure of the draw sequence.

As emphasized above, one distinct advantage of the Bayesian approach is that we can make exact finite sample inferences without resort to asymptotic

approximations. This is important only to the extent that small sample results differ from results based on asymptotics. Under very mild conditions, the posterior distribution will converge to a normal distribution centered on the true parameter value and with scale proportional to the information matrix. Inspection of the normality of various marginal posterior distributions provides a simple check on the appropriateness of large sample approximations. To examine this issue, we consider a low-dimensional choice problem with large x variation and large numbers of observations per identified parameter in order to provide a best case for normality. A three-choice model with one independent variable and three identified parameters is simulated. The true values of the identified parameters, $\tau = (\beta/\sigma_1, \sigma_{22}/\sigma_{11}, \rho) = (-2/\sqrt{2}, 4/2, 0.5)$. σ_1 denotes the standard deviation of ε_1 and ρ is the correlation coefficient. The single x variable is constructed as iid uniform on $(-0.5, 0.5)$. The parameter values and the design matrix were chosen to provide very favorable conditions for asymptotic approximations relative to what exists in many choice data sets. The x variables for different alternatives were chosen to be independent and uniform over a large range $(-0.5, 0.5)$. If we think of the independent variable as the difference of log price between two choices and the price of a third, then the relative prices fluctuate as much as 50% up and down. In addition, the slope coefficient is chosen to be -2.0 so that there is a strong 'switching' response between choice alternatives as the x variables fluctuate. In practice, many choice data sets don't display as much x variation as this simulated data and the prices may be highly intercorrelated.

Fig. 2 shows the posterior distribution of the variance ratio, σ_{22}/σ_{11} , for various simulated datasets of different sample sizes for our simple three-choice example. For each sample size, a histogram and normal quantile-quantile plot are displayed. If the marginal distribution assumed a normal shape the normal Q-Q plot should cluster very tightly around the straight line drawn in the figure. As asymptotic theory would suggest, as the sample size increases the marginal posterior distribution of the relative variances becomes more and more normal. However, it is remarkable how large a sample is required for the normal distribution to be an accurate approximation. The posterior marginal distribution is highly skewed for even as many as 500 observations per parameter. There is noticeable skewness with even 1000 observations per parameter. The fact that nonnormality arises in spite of the high degree of x variation and large N suggests that asymptotic theory may be of little use for the MNP model.

While the fact that the posterior distribution converges slowly to the asymptotic normal approximation is suggestive, this does not necessarily rule out the usefulness of asymptotic approximations for the sampling distributions of various estimators. In order to provide some insight into this issue, we conducted a sampling experiment with the simulated maximum likelihood estimator advanced by Borsch-Supan and Hajivassiliou (1990). This procedure uses the Geweke/Hajivassiliou/Keane simulation method to estimate choice

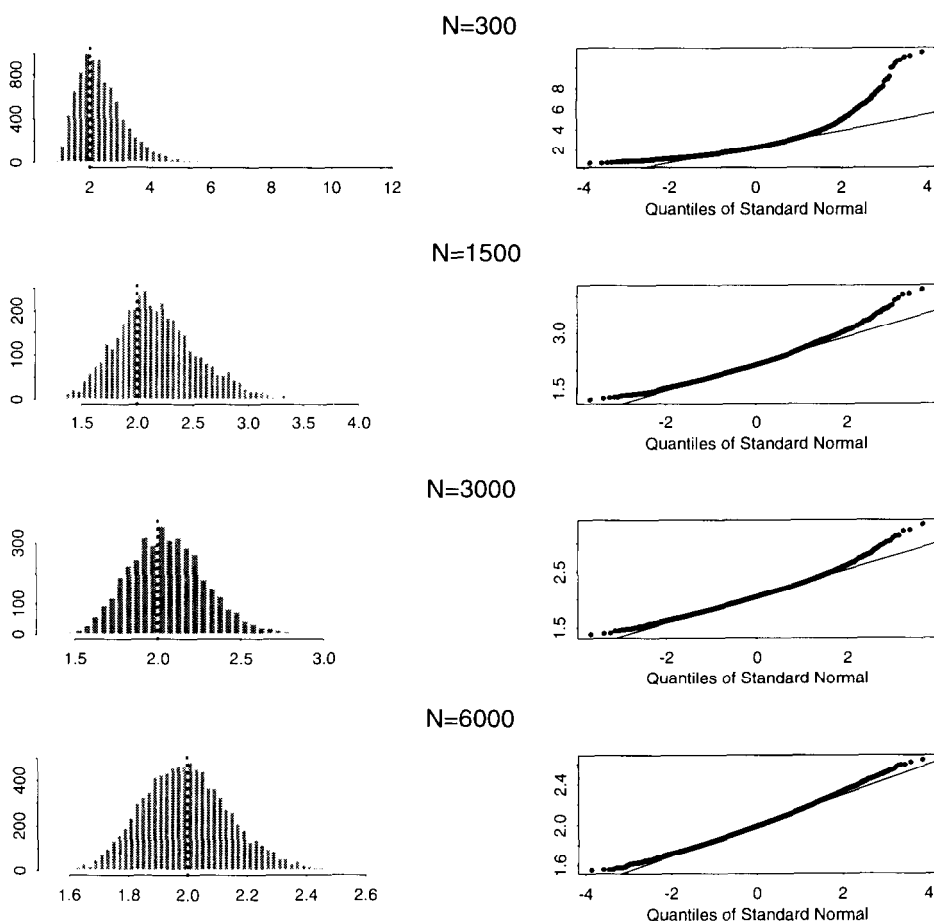


Fig. 2. Normality of posterior distribution.

probabilities and uses standard gradient methods to maximize an estimate of the likelihood formed from these estimated probabilities. For the same parameter settings and X matrices used in the three-choice posterior analysis, we simulated 500 samples of size 300, 1500, 3000, and 6000 to compute the sampling distribution of the simulated MLE. Fig. 3 reproduces the format of Fig. 2 except that the sampling distribution of the MLE of the variance ratio, σ_{22}/σ_{11} , replaces the posterior distribution. The sampling distribution of the variance ratio has pronounced skewness for as much as 1000 observations per parameter. The similarities between the posterior and sampling distributions are striking.

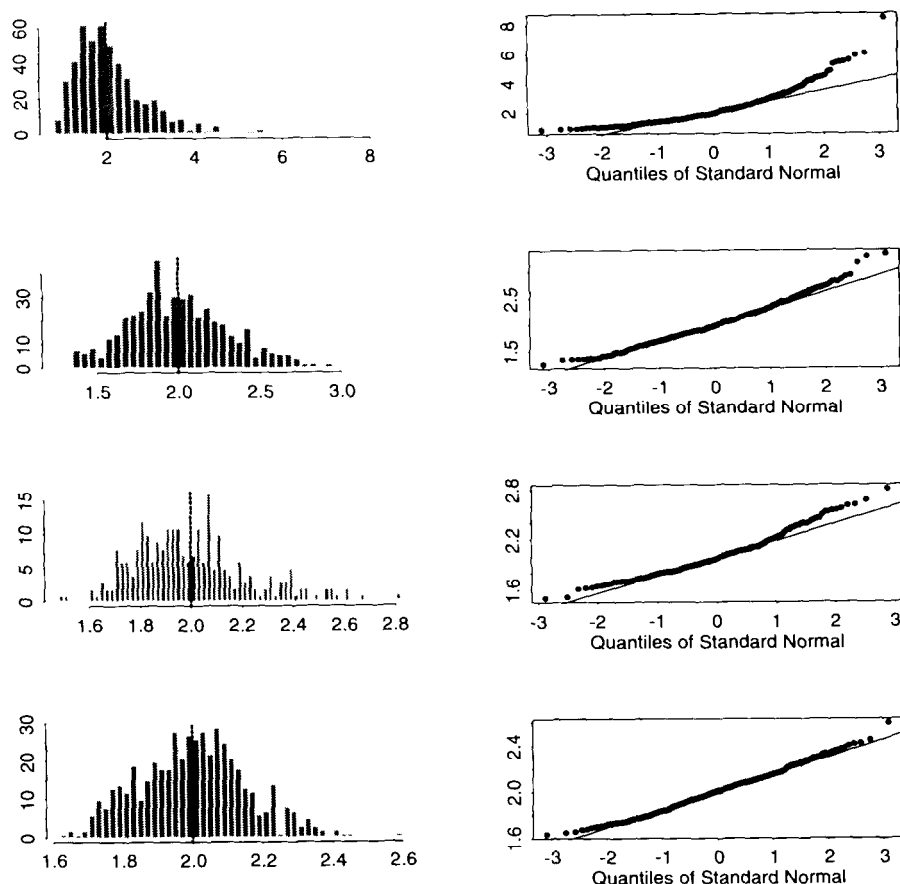


Fig. 3. Normality of sampling distribution of simulated MLE.

Quantile–quantile plots of the posterior and sampling distributions show that they are almost identical.

The adequacy of either Bayesian or sampling asymptotic approximation theory depends not only on the amount of sample information about a quantity of interest but also the nature of the parameter. For example, the regression coefficients have both posterior and sampling distributions which are well approximated by the standard asymptotic distribution theory. This is because there are no bounds on the distribution of the coefficients and there is a good deal of sample information. On the other hand, bounded functions of variance parameters such as variance ratios, correlations, and choice probabilities themselves present problems for asymptotic theory. There is relatively little

information about variance parameters even in very large samples so that the boundedness of these functions keeps the asymptotic normality from taking hold. Our approach avoids any asymptotic approximations and allows the investigator to compute an accurate approximation to the posterior distribution of any quantity of interest without resort to possibly misleading delta-method asymptotics.

The next example considered is a six-choice example with one independent variable and an equi-correlated error structure. The x values are iid draws from a uniform distribution over $(-2.0, 2.0)$. $\beta = 2$. Σ has an equi-correlated structure with large differences in relative variances,

$$\Sigma = \text{diag}(\sigma) (\rho u' + (1 - \rho) I_{p-1}) \text{diag}(\sigma), \sigma' = (5, 4, 3, 2, 1)^{0.5}, \rho = 0.5. \tag{9}$$

There are 15 identified parameters for this example: $\beta/\sigma_1, (\sigma_{ii}/\sigma_{11}, i = 1, 2, 3, 4), (\rho_{ij}, i \neq j, i = 1, 2, 3, 4)$. Table 1 shows results for the normalized slope, each relative variance, and two selected correlation coefficients; $N = 1600$. As before, all numbers in Table 1 are based on the last 10,000 draws of a 11,000 draw sequence. The 90% intervals cover each of the true values and we have the same

Table 1
A six-choice simulated example, $N = 1600$

Parameter (true value)	Prior moments		Posterior moments ^a		
	Mean	Std. dev.	Mean	Std. dev.	90% interval ^b
β/σ_1 (0.894)	0.0	12.4	0.861 (0.0051)	0.0604 (0.0048)	(0.766, 0.964)
σ_{55}/σ_{11} (0.20)	1.47	1.74	0.174 (0.0042)	0.0439 (0.0043)	(0.110, 0.256)
σ_{44}/σ_{11} (0.40)	1.51	1.92	0.486 (0.010)	0.107 (0.0090)	(0.332, 0.681)
σ_{33}/σ_{11} (0.60)	1.49	2.01	0.567 (0.011)	0.122 (0.011)	(0.389, 0.782)
σ_{22}/σ_{11} (0.80)	1.52	2.48	0.938 (0.020)	0.202 (0.018)	(0.661, 1.32)
ρ_{12} (0.5)	-0.0031	0.410	0.526 (0.010)	0.110 (0.0090)	(0.337, 0.701)
ρ_{15} (0.5)	0.00366	0.407	0.440 (0.0077)	0.0978 (0.0077)	(0.270, 0.593)

^a Standard error of posterior moment estimates from Gibbs draws in parentheses.
^b 90% posterior probability interval constructed from the 0.05 and 0.95 quantiles of the Gibbs draw distribution.

dramatic diffuseness of the induced prior and the tight posterior. As before, the marginal posterior distributions of both the relative variances and the correlation coefficients are very nonnormal. The degree of autocorrelation in the Gibbs sequence is high for $p = 6$. A lag of 200 was used to compute the Newey–West standard errors. At this level of autocorrelation, the Gibbs draws are approximately 1/10th as informative as an iid sample, if indeed it were possible to sample iid from the posterior distribution. However, we still find the Gibbs sequence to be sufficiently accurate for most uses. In order to make reliable inferences, it is important that the sampling variability in the estimates based on Gibbs draws be small relative to the posterior variation.

As discussed in Section 4, our solution to the identification problem is to specify diffuse but proper priors over the parameter space and marginalize on identified functions of the parameters. Table 1 shows the induced prior distribution over selected identified parameters including normalized slope coefficient, variance ratios, and correlation coefficients. Comparison of the prior and posterior distributions in Table 1 shows that our calibration of the prior is very diffuse so that the posterior distribution reflects sample and not prior information. An alternative way of verifying the relative effect of prior and sample information is to perform a prior sensitivity analysis in which the prior diffusion is varied and changes in the posterior are recorded. Fig. 4 shows the results of our prior sensitivity experiment in which we systematically varied the joint prior over β and Σ . For the case of β , we doubled and halved the diffusion of the normal prior on β , setting $A = 1/4A^*$ and $A = 4A^*$ where A^* is the base precision in the base case reported in Table 1. For the prior on Σ^{-1} , we increased the degrees of freedom parameter, ν , from 6 to 12 which has the effect of tightening the prior over the location I_{p-1} . Note that it is not possible to have a more diffuse prior on Σ since $\nu = p$ is the smallest value for which the prior will remain proper. Thus, Fig. 4 shows four priors: the base case, a tighter prior on Σ , a tighter prior on β , and a more diffuse prior on β . The three left-hand panels of Fig. 3 display the prior densities for β/σ_1 , σ_{55}/σ_{11} , and ρ_{12} , while the right-hand graphs show the corresponding posterior densities. Each of the four priors/posteriors are displayed on the same panel with different line types and shading. With even large changes in prior diffusion, the posterior distributions remain virtually unchanged.

In previous sections, we have stressed the practical importance of determining that the effects of initial conditions are dissipated before the Gibbs draws are judged to have converged to a draw from the stationary distribution. In order to check this, we started the MNP sampler from nine different starting conditions. For each of the nine start points, we drew 5000 Gibbs draws and plotted the marginal distribution of the last 4000 draws. Our nine start points are arranged in a two-factor orthogonal design.

$$\begin{aligned}\theta_{ij} &= (\beta_i, \Sigma_j), \quad i, j = 1, 2, 3, \\ \beta_i &= (9, 2, -5),\end{aligned}\tag{10}$$

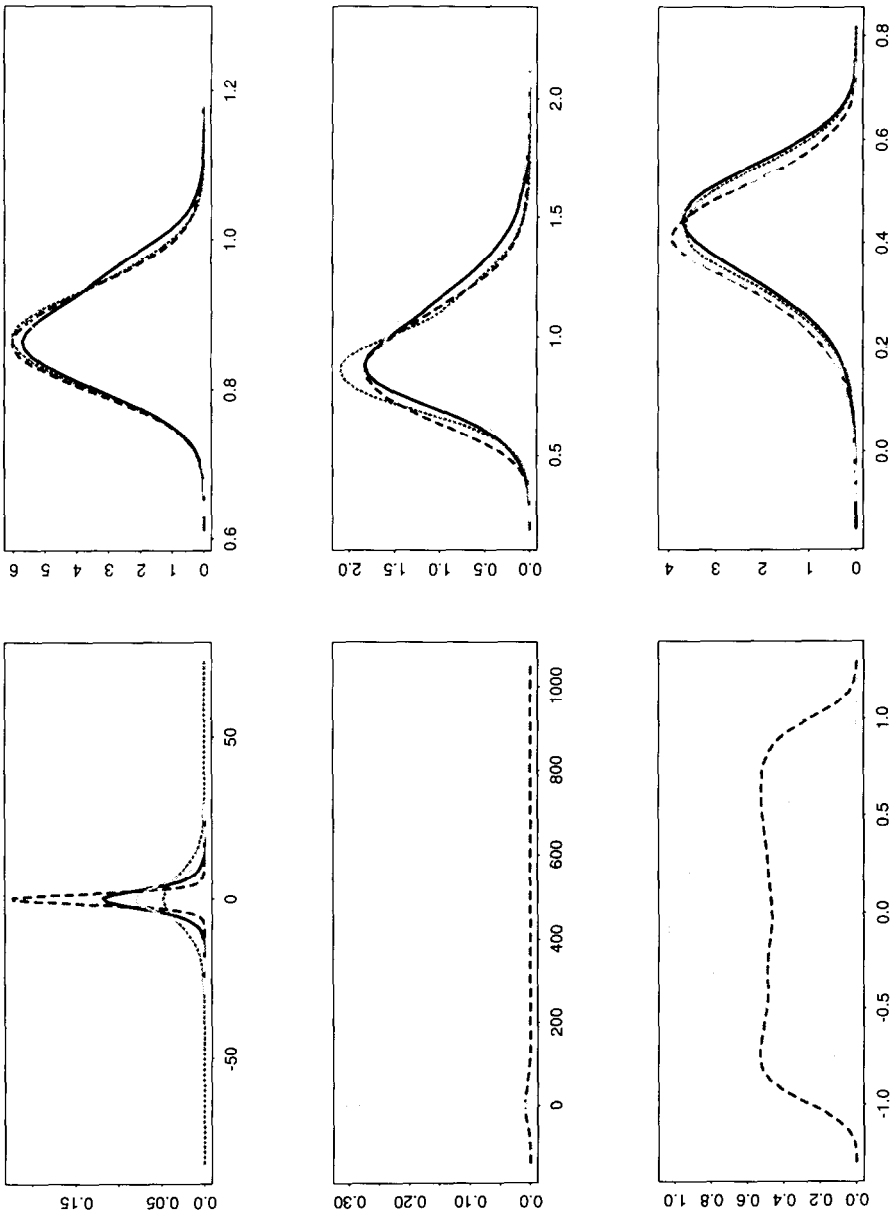


Fig. 4. Prior sensitivity analysis.

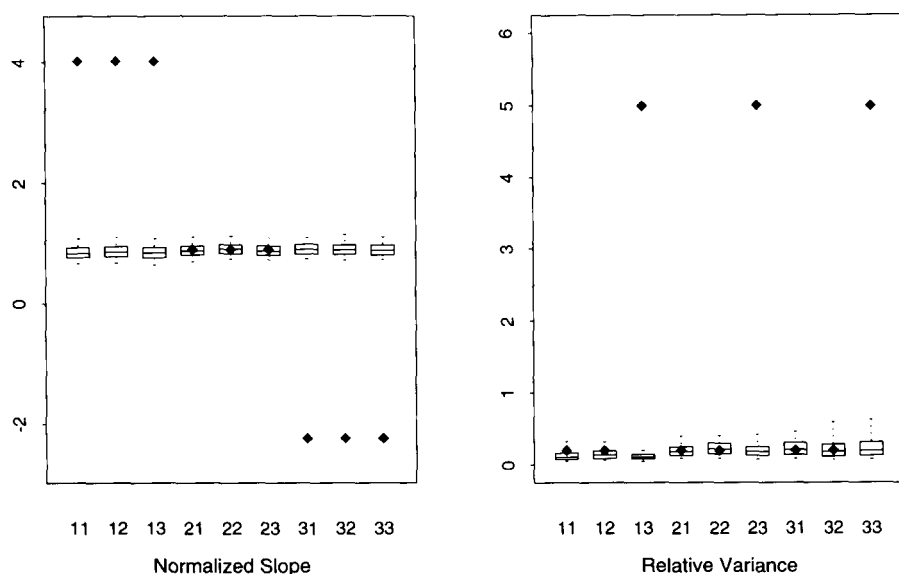


Fig. 5. Sensitivity to initial conditions.

$$\Sigma_1 = \text{diag}(\sigma) (0.9I' + 0.1I_5) \text{diag}(\sigma), \quad \text{where } \sigma' = (5, 4, 3, 2, 1)^{0.5},$$

$$\Sigma_2 = \text{diag}(\sigma) (-0.2I' + 1.2I_5) \text{diag}(\sigma), \quad \text{where } \sigma' = (5, 4, 3, 2, 1)^{0.5},$$

$$\Sigma_3 = \text{diag}(\sigma) (0.5I' + 0.5I_5) \text{diag}(\sigma), \quad \text{where } \sigma' = (1, 2, 3, 4, 5)^{0.5}.$$

Figs. 5 and 6 show boxplots of the estimated posterior marginals for each of the nine sequences of draws. The center line in the box is the median and the height of the box is the distance between the 0.1 and 0.9 quantiles; the 'whiskers' extending above and below the box represent the 0.01 and 0.99 quantiles. The diamond-shaped black dots represent the starting point for the normalized parameter. Fig. 5 presents the sensitivity analysis for the normalized slope and σ_{55}/σ_{11} while Fig. 6 shows the analysis for two correlation coefficients. Relative to the starting point variation, there are few differences between the resulting marginal distributions. There is some variation in the ρ_{15} distributions. The high degree of autocorrelation in the Gibbs sequences and the lack of a systematic relationship between the starting point and the resulting marginal suggest that these differences are due to sampling error in the estimates of the distribution and not to sensitivity to initial conditions.

Examining posterior marginals provides only partial information about the joint posterior distribution. Table 2 provides the posterior correlation matrix of model parameters for selected parameters in the six-choice case. There is

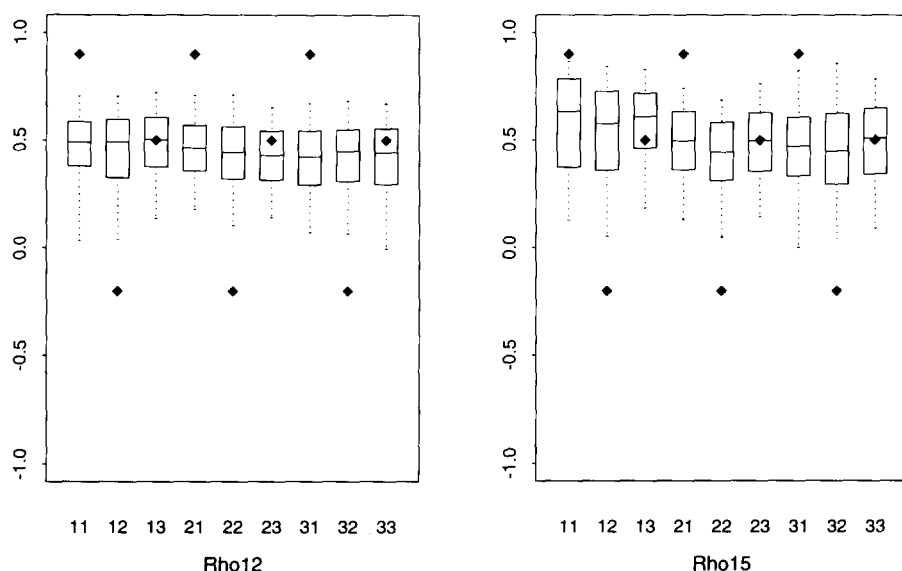


Fig. 6. Sensitivity to initial conditions.

Table 2

Posterior correlations, six-choice simulate example

β/σ_{11}	σ_{55}/σ_{11}	σ_{44}/σ_{11}	σ_{33}/σ_{11}	σ_{22}/σ_{11}	ρ_{12}	ρ_{15}
1.00	0.45	0.66	0.63	0.71	-0.42	-0.35
	1.00	0.34	0.49	0.42	-0.19	0.001
		1.00	0.20	0.46	-0.33	-0.14
			1.00	0.36	-0.36	-0.31
				1.00	-0.29	-0.33
					1.00	0.23
						1.00

a reasonably high degree of correlation, particularly between the normalized intercept and the relative variance parameters.

7. Computational issues

The practical usefulness of our MNP sampler depends critically on computational feasibility. Our simulated examples illustrate that in order to assure

convergence and obtain precise estimates of the posterior distributions large numbers of draws will be required (on the order of 10,000 draws for six-dimensional systems). In this section, we provide an explicit formula for the number of floating point operations required per MNP draw and conduct a timing exercise to verify the formula and provide a means of predicting run times for other problems.

The number of floating point operations and univariate random variate draws suggests that the time to complete one MNP draw should be a polynomial in N, p, k with important terms $N(p-1)$, $N(p-1)^2$, $N(p-1)k^2$, and k^3 . In order to calculate the coefficients, we timed a series of MNP sampling runs conducted on an orthogonal array of N, p , and k values. A full factorial design with $N = (100, 400, 800)$, $(p-1) = (2, 4, 8)$, and $k = (p, p+1)$ was used. The following regression summarizes the results:

$$\begin{aligned} \text{Secs} = & 0.63 + 0.0081N(p-1) + 0.00024N(p-1)^2 \\ & + 0.000074N(p-1)k^2 + 0.0059k^3, \end{aligned}$$

Observations = 18,

Standard error of the regression = 0.9, $R^2 = 0.99$.

Secs is the number of CPU seconds³ for 100 draws. Using this regression, we would forecast that a modest problem with 1000 observations, 6 choices, and 5 independent variables would run at the rate of 9.5 minutes per 1000 draws. A more ambitious problem with 5000 observations, 11 choices, and 10 independent variables would run at the rate of 150 minutes/1000 draws.

8. A Bayesian random coefficient approach and panel data example

8.1. Bayesian hierarchical models

Since it is rarely possible to observe the same economic agent on many choice occasions, panel data on consumer choice is often used for applications of multinomial choice models. In the panel setting, it is important that the model accommodate heterogeneity. The standard approach to modelling consumer heterogeneity in panels with small T and large N is a random coefficients model (see Heckman, 1982, for example). Hausman and Wise (1978) motivate the MNP

³ All computations are performed on SUN Microsystems Sparc 2 computers which are rated at 27 MIPS and 4.0 MFLOPS double precision running SUNOS 4.1.2 and using SUN fortran 1.4 at the highest level of optimization. NAG Mark 14 algorithms are used to make the basic univariate random number draws.

model through a random coefficients approach in which variation in the regression coefficients induces a special correlation structure. The multivariate regression latent variable structure is ideally suited for a Bayesian hierarchical model. In linear models, a Bayesian hierarchical approach was pioneered by Lindley and Smith (1972). Gelfand et al. (1990) demonstrate how the Gibbs sampler is ideally suited for analysis of linear random coefficient models (see also Morris, 1987).

Consider the latent variable structure for a panel of N households observed over T_i periods,

$$w_{it} = X_{it}\beta_i + \varepsilon_{it}, \quad i = 1, \dots, N, \quad t = 1, \dots, T_i, \quad (11)$$

$$\varepsilon_{it} \sim \text{iid } N(0, \Sigma).$$

In the hierarchical model, we assume that each β_i is drawn from a superpopulation. For convenience, we assume that this superpopulation is normal. $\beta_i \sim \text{iid } N(\bar{\beta}, V_\beta)$. It is also possible to replace this normal prior in the first stage of the hierarchical model by a finite mixture of normals.

$$p(\beta_i | \lambda, \{\bar{\beta}_j\}, \{V_{\beta j}\}) = \sum \lambda_j \phi(\beta_i | \bar{\beta}_j, V_{\beta j}), \quad \sum \lambda_j = 1, \quad j = 1, \dots, s. \quad (12)$$

We would have to introduce a prior for the λ_j mixture probabilities, and it would be most convenient to use a standard natural conjugate Dirichlet prior. The mixture of normals model could easily be handled in the Gibbs sampler via the introduction of latent indicator variables which switch on for each of the s normal components in the mixture.

In the random coefficient literature, it is common to assume that there are random effects at the household level and these are often modelled by random household intercepts (cf. Borsch-Supan and Hajivassiliou, 1992). If the random intercepts are normally distributed across households, then this introduces a complicated $T_i \times p$ correlation structure across all observations for the same household. In principle, this approach could be generalized to our hierarchical model in which the entire regression coefficient vector is random across households, although this has not yet been implemented in the literature. If the distribution of the regression coefficient vector is nonnormal and continuous (as in the mixture of normals given above), then standard random coefficient approaches would require numerical integration methods to average the household likelihood conditional on β_i over the nonnormal distribution.

Once the random coefficient source of heterogeneity is properly modelled, it might be the case that there is little correlation remaining in the error terms, ε_{it} , and that one should specify an independence probit in which Σ is diagonal. Our approach is to specify a full covariance probit and use the data to infer about the extent to which Σ is diagonal.

In the Bayesian approach to hierarchical models, an additional third stage is added to the model in which we specify priors over $\bar{\beta}$ and V_{β} . For convenience and flexibility, we specify a normal prior for $\bar{\beta}$ and a Wishart prior for $H = V_{\beta}^{-1}$. $\bar{\beta}$ and H are assumed to be a priori independent,

$$\bar{\beta} \sim N(\mu_{\beta\text{bar}}, V_{\beta\text{bar}}) \quad (13)$$

and

$$H = V_{\beta}^{-1} \sim \text{Wishart}(v_0, V_0). \quad (14)$$

In many applications, the prior on $\bar{\beta}$ is taken to be improper or, at least, very diffuse. We will follow this practice but keep the prior proper so as to ‘contain’ the Gibbs sampler (see the section on identification above). This prior specifies that the $\{\beta_i\}$ come from some common distribution whose mean is not known, a priori, but with a restricted amount of dispersion. The key parameters of this prior are v_0 and V_0 which determine the values of V_{β} on which the prior places mass.

Our MNP sampler can easily be adapted to handle the hierarchical model by appending the variables $\{\beta_i\}$, $\bar{\beta}$, V_{β} to the set of variables we draw from. The modified MNP sampler would draw in turn from five sets of conditional distributions:

- 1) $w_{itj} | w_{it-j}, \{\beta_i\}, G, \bar{\beta}, H, d, \quad i = 1, \dots, N, \quad t = 1, \dots, T_i, \quad j = 1, \dots, p-1,$ (15)
- 2) $\beta_i | w, G, \bar{\beta}, H, d,$
- 3) $G | w, \{\beta_i\}, \bar{\beta}, H, d,$
- 4) $\bar{\beta} | w, \{\beta_i\}, G, H, d,$
- 5) $H | w, \{\beta_i\}, G, \bar{\beta}, d.$

Each of these groups of conditional distributions have simple forms which can be easily drawn from. Conditionals 4) and 5) which reflect the hierarchy are the only new sets of conditionals. As is well-known, 4) is in the multivariate normal form and 5) is a Wishart ($v_0 + N, (V_0 + \Sigma(\beta_i - \bar{\beta})(\beta_i - \bar{\beta})')$).

In some problems, it may be useful to make inferences about the draw of β for household i . In the Bayesian hierarchical approach, we can compute, at least in principle, the posterior distribution of the draw,

$$p(\beta_i | \text{data}) = \int p(\beta_i, \bar{\beta}, V_{\beta}, \Sigma | \text{data}) d\bar{\beta} dV_{\beta} d\Sigma. \quad (16)$$

The sequence of Gibbs draws of $\{\beta_i\}$ can be used to make posterior inferences about particular panel members. These draws are available ‘free’ for the cost of

storage during the normal course of running our MNP sampler for the hierarchical model. Thus, the Bayesian approach combines the smoothing advantages of the frequentist random effects model with the richness of the fixed effects approach. Of course, in the Bayesian point of view there is no distinction between fixed and random effects models, only between hierarchical and non-hierarchical models.

8.2. *A panel data example*

Widespread diffusion of optical scanning, point-of-purchase equipment in the grocery industry has made it feasible to collect detailed and highly accurate consumer panel data. The largest two marketing research firms, A.C. Nielsen and Information Resources Inc., have developed a number of household panels starting in the early 1980's. Our panel data comes from the ERIM panels maintained by A.C. Nielsen in Springfield, MO and Sioux Falls, SD in the period 1985–1988. Nielsen obtained the cooperation of all supermarkets in these small cities and subsidized the installation of optical scanning equipment where necessary. Households were invited to enroll in the panel and given various incentives which mostly consisted of eligibility for various prizes. At each supermarket visit, the household would present its bar-coded membership card and a complete record of household purchases would be made and stored for later retrieval. Once a week, price and display information was downloaded from the stores point-of-sale computers. Upon enrollment in the panel, each household was surveyed to collect various demographic information and information on household production technology such as appliance ownership. In addition, for a subsample of households, detailed television viewing information including exposure to specific advertisements was collected from monitoring devices attached to individual televisions. The commercial purpose of these panels was to conduct experiments involving new ads and new products.

In comparison to panel data employed in much empirical work in economics, these panels are more detailed and arguably freer from measurement error due to the use of optical scanning methods. However, the detail and sheer size of the datasets provide a challenge to the modeler. Purchases are recorded for each Uniform Product Code (UPC); UPCs are assigned to each unit brand, flavor, packaging, and size of a product. As there are some 35,000 to 50,000 UPCs in a typical supermarket, the analyst must either restrict attention to a subset of UPCs or aggregate the data. Furthermore, A.C. Nielsen has chosen to release data in only a small subset of product categories rather than the full set of purchase records which means that information on some substitutes or complements for a given product category is often unavailable. Finally, the voluntary nature of the panel opens the possibility that there is selection bias and attrition bias.

Our approach here is to build a simple choice model for consumer purchases in the margarine product category both to illustrate the usefulness of our MNP sampler as well as this value of this relatively unexplored source of data.⁴ Our sample of 150 households have 2236 purchase records for six top-selling brands. These six top brands account for more than 70% of all purchases by all households in the margarine category.

The brands used in the analysis, sample frequency of purchase, and price averaged over the 2236 purchase occasions are given below:

Brand name	Purchase frequency	Average price
Parkay	0.43	61 ¢
BlueBonnet	0.070	64 ¢
Fleischmann's	0.060	106 ¢
House Brand	0.20	53 ¢
Generic Brand	0.15	44 ¢
Shedd Spread Tub	0.09	80 ¢

We propose a simple version of the MNP in which there are six regressors: five brand-specific intercepts and the log of price. Our reference brand is Parkay. The error terms follow a general five-dimensional normal distribution. High variability in relative prices and low intercorrelation make this data very different from many datasets used in the applied demand literature.

This simple MNP specification can be derived from a utility structure in which the vector of consumption of the margarine brands is weakly separable from the consumptions of other goods. $U(z, x) = v(z, g(x))$, where x is the vector of margarine consumption and, for convenience, z is a composite 'outside' good. As is standard in the random utility literature, if we assume that $g(x) = \psi'x$ and that the marginal utility is composed of a fixed component with multiplicative log-normal errors, the derived demand system conditional on a purchase is in the form of the simple MNP specification we have chosen. A full unconditional demand analysis requires the specification of the bivariate utility function, $v(\cdot, \cdot)$ (see, for example, Hanemann, 1984) and requires total grocery expenditure data and price information for products outside the margarine category. Given our incomplete information on other product categories, we decided not to pursue this unconditional specification.

We assume that the purchase observations are conditionally independent for each household. Margarine is a frequently purchased, low-cost item and there is

⁴ERIM academic datasets are available for a nominal fee from the Marketing Science Institute, Cambridge, MA.

Table 3

A six-brand hierarchical choice model: Posterior moments of selected parameters, $N = 2236$

Parameter	$\bar{\beta}$	Std. dev. across HHs ^a
<i>Brand intercepts</i>		
Blue Bonnet	– 1.75 (0.22)	1.46 (0.19)
Fleischmann's	– 3.46 (0.77)	3.94 (0.67)
House	– 3.86 (0.74)	4.57 (0.70)
Generic	– 6.45 (1.13)	4.92 (0.94)
Shed	– 2.42 (0.67)	2.99 (0.61)
Price coefficient	– 5.23 (0.72)	3.08 (0.52)

^a Square root of diagonal elements of V_{β}/σ_{11} .
Posterior means (standard deviations).

little evidence in the data of 'stock-piling' which is our justification for imposing this assumption. A full dynamic economic model involving consumer expectations of future price changes and inventory behavior is beyond the scope of this analysis.

Even casual inspection of this sort of panel data indicates that households are very heterogeneous in their choice behavior. Some households are observed to purchase mostly the lowest-price brand, while others are very 'loyal' to other brands. In order to accommodate this sort of heterogeneity, we employ the Bayesian random coefficient or hierarchical model outlined above in which each household's β vector is viewed as a draw from a normal superpopulation with mean $\bar{\beta}$ and variance V_{β} . There are three key priors for this model: the prior on Σ , the prior on $\bar{\beta}$, and the prior on V_{β} . We assume relatively diffuse priors for each of these quantities. $\Sigma^{-1} \sim W(p+4, (p+4)I_p)$, $\bar{\beta} \sim N(0, 100I_k)$, $V_{\beta}^{-1} \sim W(k+4, (k+4)I_k)$. The marginal distribution of the identified parameters induced by these priors is also very diffuse.

Table 3 presents the results of the application of our random coefficient MNP sampler. The first column presents the posterior means and standard deviations of elements of the $\bar{\beta}/\sigma_1$ vector. As might be expected in a choice situation with close substitutes, the average price coefficient is very large (–5.2). The generic stick margarine has the largest negative intercept which indicates that, on average, households accord a much lower marginal utility to the generic brand.

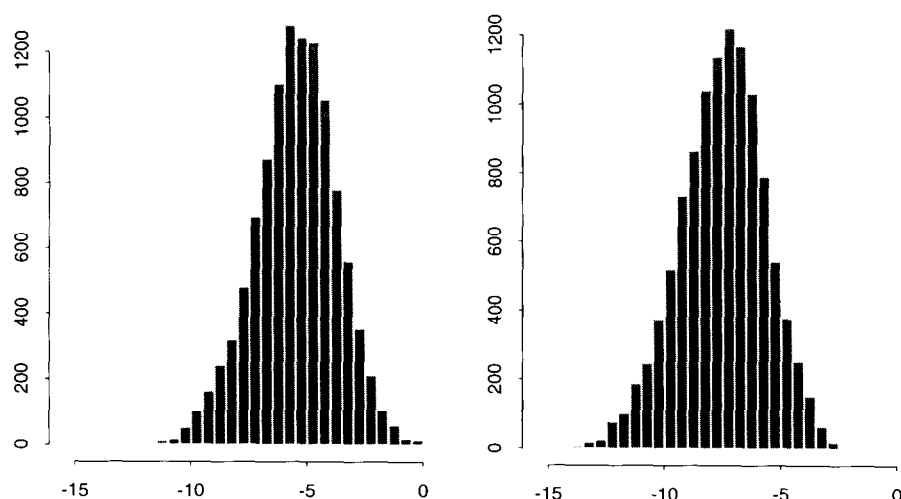


Fig. 7. Household price coefficient distributions.

By examining the posterior distribution of suitably normalized elements of V_β , we can gauge the degree of household heterogeneity. The second column shows the posterior means and standard deviations of the square root of the diagonal elements of V_β/σ_{11} . For example, while the mean of the price coefficients across households is estimated to be -5.2 (using the posterior mean as a Bayes estimate), the standard deviation across households is huge at 3.1 . In addition, there is even wider dispersion in the marginal utility accorded to the house and generic stick brands. All calculations in Table 3 are based on 11,000 draws extracted from the end of a 15,000 draw sequence; all numerical standard errors are below 0.001 .

One distinct advantage of the Bayesian hierarchical approach is that, in principle, it is possible to make inferences about the draws from the superpopulation for a given household. Computing this posterior distribution involves a high-dimensional integral which has kept investigators from computing the exact posterior. Instead, many rely on approximations which simply 'plug-in' the MLEs of $\bar{\beta}$ and V_β . By saving and using the β_i draws from our MNP sampler we can compute the exact marginal distribution of the coefficient vector on a household by household basis. Fig. 7 shows the marginal distributions of the price coefficients for two households with about 30 observations per household. The household corresponding to the distribution on the right appears to be more price-sensitive than the left household.

Fig. 8 shows the marginal posterior distributions of the correlation coefficients of the random utility errors between brands. The brands are ordered as BlueBonnet, Fleischmann's, house, generic, and Shed tub with corresponding

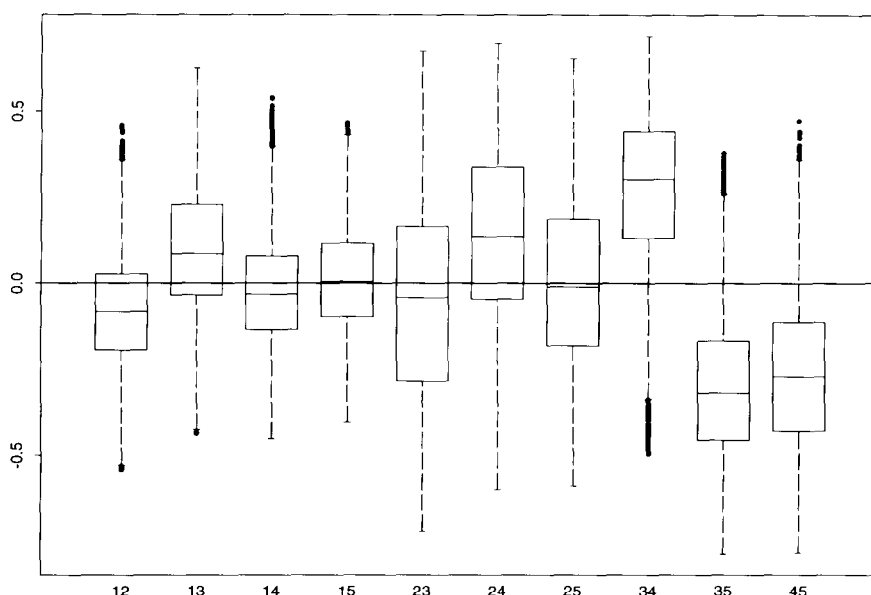


Fig. 8. Posterior distributions of correlations.

subscripts 1, ..., 5. Three of the ten correlations have posterior distributions with mass well away from zero. The correlation between the house and generic random errors is positive while the house/Shed and generic/Shed correlations are negative. Thus, the evidence in our data does not favor an independence probit structure. In general, there is a good deal of dispersion in these correlation distributions indicating how difficult it is to infer about the correlation structure.

9. Extensions of the model

Our new approach to the analysis of the MNP lends itself naturally to a wide variety of important extensions. In this section, we discuss two important new extensions of the basic MNP model which can be analyzed with variants of our MNP sampler: 1) Variance components approaches can be used to give structure to the correlation matrix. 2) Multiperiod probit models can be easily handled.

9.1. Variance component approaches

One of the difficulties with the MNP model is that the number of error covariance parameters expands dramatically as the number of choice alternatives

increases. One solution to this problem is to impose restrictions on the covariance matrix such as an equi-correlated or patterned covariance structure. The nested logit model proposed by McFadden (1977) can be viewed in this vein. Another important variance component structure arises in a multiperiod probit model in which there is a common aggregate shock which influences all households in addition to household specific shocks. With iid aggregate shocks, this will induce an equi-correlated correlation matrix across discrete events for the same household. Although it is difficult to impose exact restrictions on Σ in our MNP sampler approach, we can handle this problem easily by the use of variance components. Consider a variance components set-up with G groupings of choice alternatives,

$$\varepsilon_{ij} = z_{ij} + v_{ig}, \quad j = 1, \dots, p-1, \quad g = 1, \dots, G, \quad (17)$$

$$z_{ij} \sim \text{iid } N(0, \sigma_z),$$

$$v'_i = (v_{i1}, \dots, v_{iG}) \sim \text{iid } N(0, A).$$

To implement the sampler, we add the variance components themselves to the list of sample random variables, dropping the full Σ array. Inference about the new variance parameters, σ_z and A , can be accomplished through the usual sequence of conditional draws. The key to the successful implementation of a Gibbs strategy for variance components is the recognition that posterior inference about $v_{ig}|v_{i-g}, \varepsilon_i$ is a simple normal mean problem.

9.2. Multiperiod probit models

In many panel applications of choice models, the assumption of independence of the choice model errors across time for the same household is questionable. One common view is that there are aggregate unobservable factors common to all households in the panel as well as household-specific unobservable factors. This will introduce autocorrelation into the error structure if the aggregate disturbances are autocorrelated.⁵ It is straightforward to extend our MNP sampler to handle error terms which follow an $AR(p)$ process. To illustrate this idea without introducing excessive notation, we will consider a binomial

⁵ A simple component structure, $\varepsilon_{it} = v_{it} + u_t$ with v_{it} iid and u_t following an $AR(p)$ process would induce an $ARMA(p, 1)$ structure for the ε_{it} errors. To simplify the exposition, in this section we specify an $AR(p)$ process directly for the error terms. As is well known, $AR(p)$ processes can well approximate $ARMA$ processes as long as p is reasonably large. It is always possible to put an informative 'smoothness' prior on the AR coefficients. If an $ARMA(p, q)$ process is desirable, our MNP sampler can be modified using a hybrid of Gibbs and Metropolis methods to handle the MA terms (see Tierney, 1991, for a discussion of hybrid Markov chain samplers and Chib and Greenberg, 1992, for analysis of linear regression models with $ARMA$ errors).

multiperiod probit model. All of the ideas extend without modification to the multinomial case.

The sample of N households which follow a binary multiperiod probit model for T_i periods. The latent variable set-up for this problem is

$$w_{it} = x'_{it}\beta + \varepsilon_{it} \quad \text{where} \quad \varepsilon_{it} = \Gamma_p(B)\varepsilon_{it} + u_{it}, \quad \sigma_u^2 = 1. \quad (18)$$

The $AR(p)$ polynomial is parameterized by p autoregressive coefficients which we denote by the vector ϕ . The u_{it} are assumed independent across time and households. To implement the Gibbs sampler, we must slightly modify our strategy for drawing w and append two sets of conditional draws to set of Gibbs draws. The Gibbs strategy consists of the following sets of conditionals:

$$\begin{aligned} w &| \beta, \phi, \varepsilon_0, d, \\ \beta &| w, \phi, \varepsilon_0, d, \\ \varepsilon_0 &| \phi, w, \beta, d, \\ \phi &| \varepsilon_0, w, \beta, d. \end{aligned} \quad (19)$$

Here ε_0 is the vector of $N \times p$ initial errors for each of the N households. The draw of w proceeds using the same strategy as before except that the autocorrelation structure changes the univariate normal conditional distribution of $w_{it}|w_{i-t}$. In addition, the conditional posterior of β is computed as before except on the orthogonalized regression system (premultiply by the Cholesky root of the correlation structure matrix which is available since we are conditioning on ϕ). Inclusion of the Gibbs step involving ε_0 allows us to base our inferences on the unconditional likelihood rather than conditioning on the first p observations. In panel applications, this can be important since we often observe households in only a handful of periods.

Appendix: Geometric convergence for the binomial case

In this appendix we prove geometric ergodicity for MNP samplers which rely on direct sampling from the truncated normal distribution of $w_i|d_i$. For the binomial probit case, these are univariate truncated normal draws which can be made very efficiently using the accept/reject method outlined in Section 3 above. In the multinomial case, efficient accept/reject methods are not available and we must 'Gibbs' through the sequence of conditional truncated variables as discussed in Section 3.

We require a modest investment in notation. Let $x = (w, \beta, \text{vec}(G))$ denote an element of E , the *state space* of the Markov chain defined by the Gibbs sampler. $E = \{W_y \times \Omega\}$. $W_y \subset R^{(p-1)N}$. The W_y is the region defined by the truncation induced by the observed choices, y . W_y is the cartesian product of

$N(p-1)$ -dimensional cones in R^{p-1} . Note that a Bayesian analysis proceeds conditional on the observed data. $\Omega \subseteq \{(\beta, \text{vec}(G)) \text{ such that } \beta \in R^k \text{ and } G > 0 \text{ (space of all positive definite matrices)}\}$. $\mathcal{E}$ is the usual σ -algebra on E . The Markov transition kernel of the chain on $(E, \mathcal{E})$ is a map $P: E \times \mathcal{E} \rightarrow [0, 1]$ such that (i) for any $A \in \mathcal{E}$, $P(\cdot, A)$ is measurable, and (ii) for any $x \in E$, $P(x, \cdot)$ is a probability measure of $(E, \mathcal{E})$. For a fixed x , P acts as a conditional probability measure.

An geometrically ergodic Markov chain with invariant distribution π requires the existence of a nonnegative function M with finite π expectation and positive constant $r < 1$ such that $\|k^n(x, \cdot) - \pi\| \leq M(x)r^n$. Here k^n is the transition kernel of the chain after n steps and $\|\cdot\|$ is the total variation norm.

If the state space is not bounded, it is difficult to directly verify geometric or uniform ergodicity. To develop sufficient conditions for geometric/uniform ergodicity, we must introduce the concepts of a minorization condition and a small set. A π -irreducible kernel P satisfies a *minorization condition*, $M(m, \beta, C, v)$, for an integer $m \geq 1$, a constant $\beta > 0$, a set $C \in \mathcal{E}$, and a probability measure v on $\mathcal{E}$ if $\pi(C) > 0$ and $\beta v(\cdot) \leq P^m(x, \cdot)$ for all $x \in C$. A set C is a *small set* for P if P satisfies the minorization condition $M(m, \beta, C, v)$ for some m, β , and v .

A sufficient condition for geometric ergodicity is given by the following proposition (see Numelin, 1984, Proposition 5.21; Tierney, 1991, Proposition 1):

Proposition. Suppose X_n is ergodic and $\exists$ a nonnegative, $\mathcal{E}$ -measurable function g , a small set C , and a constant $r > 1$ such that

$$\sup_{x \in C} E[rg(X_{n+1}) - g(X_n) | X_n = x] < 0$$

and

$$\sup_{x \in C} E[g(X_{n+1}); X_{n+1} \in C^c | X_n = x] < \infty,$$

then X_n is geometrically ergodic.

The intuition behind the two conditions in the Proposition can be established by thinking of C as some bounded set and of g as a function which increases off of the bounded set. The first condition says that if the chain is at a state outside of C at step n , then after one more step the chain will be ‘attracted’ to the set C in the sense that, given a starting point outside of C , the chain will move with high probability to points with lower g values. Because g increases away from C , points with lower g values are ‘closer’. The second condition says that if the chain starts in C , then you can’t drift too far from C in the sense that the supremum of the conditional expectation of g must be finite.

For the case in which direct draws are available from the truncated normal distribution, we can directly verify geometric ergodicity using the Proposition with only one innocuous assumption that the regression coefficients are in a bounded set and the inverse of the variance-covariances is bounded away from zero.

Theorem. Let $\Omega = \{(\beta, \text{vec}(G): \beta \in B \text{ and } \text{vec}(G) \in \mathcal{G})\}$. If B is a compact subset of R^k and $\mathcal{G}$ is a compact subset of the set of positive definite matrices, bounded away from zero, then the MNP Gibbs sampler satisfies the drift condition in the Proposition and is, therefore, geometrically ergodic. *pf.*

Consider w_i , the latent variable vector for the i th observation. Since the observations are iid, we can treat each observation independently of the others in the Gibbs sampler,

$$w_i | \beta, G \sim N(X_i \beta, G).$$

Given y_i , we truncate the distribution of w_i to a cone in R^{p-1} . If $y_i = j$, the cone is defined by $W_{yi} = \{w: w_j \geq \max(w_{-j}, 0)\}$.

Let $g(x) = w_1^2$, that is $g(\cdot)$ serves to pick off one element of the w vector and g is large away from zero.

$E[g(x) | w', \beta', G', y] = f(\beta', G', y)$. Note that y remains fixed in the Bayesian analysis and the truncation point does not depend on any element of the w vector.

Since β', G' are in a bounded set, f is bounded above by $\bar{f}$.

If we take the set C to be a bounded set but allow for large enough values of w , then the Proposition will be satisfied.

$$\sup_{x \in C'} E[rg(X_{n+1}) - g(X_n) | X_n = x] = \sup_{(\beta', G') \in C'} E[rf(\beta', G') - w_1'^2 | \beta', G']$$

$$< r\bar{f} - \inf_{w_1 \in C'} w_1'^2.$$

By taking C large enough, we can insure that the above expression is < 0 for $r > 1$.

Finally, any bounded C is a small set since C satisfies a minorization condition with $m = 1$ and ν taken to be a measure with density given by the product of the conditionals, $p^*(w | \theta', y)$ and $p(\theta | w', y)$.

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